

Matrix Multiplication using

Systolic Arrays

ES 204 : Digital Systems
Project

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Problem Statement

- Implementation of 3 x 3 Matrix multiplication using systolic array and implementation of UART to operate these modules using a python script on a computer.

3 x 3 Matrix

$$\bullet A = \begin{matrix} a_{00} & a_{01} & a_{02} \\ a_{10} & a_{11} & a_{12} \\ a_{20} & a_{21} & a_{22} \end{matrix}$$

$$\bullet B = \begin{matrix} b_{00} & b_{01} & b_{02} \\ b_{10} & b_{11} & b_{12} \\ b_{20} & b_{21} & b_{22} \end{matrix}$$

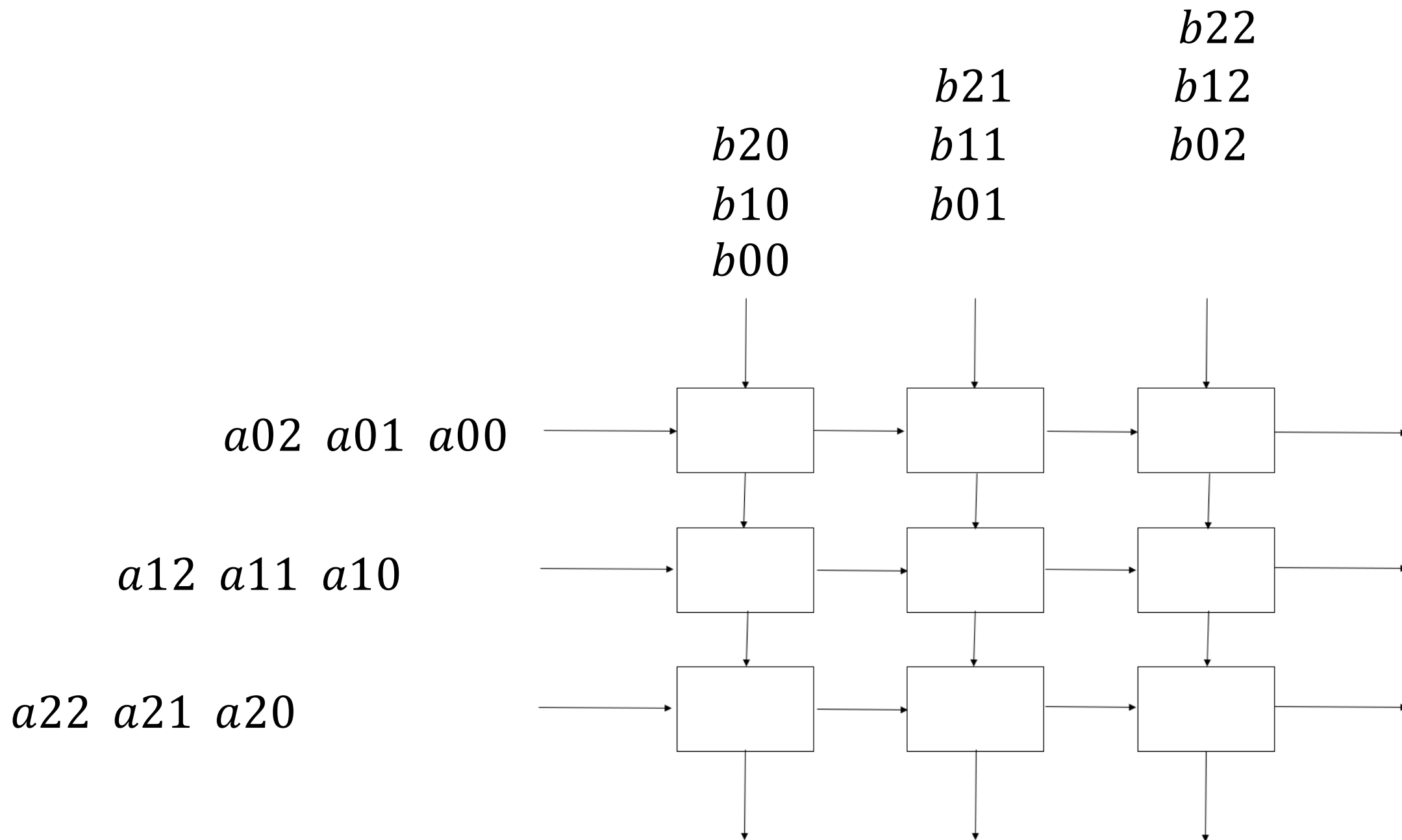
$$time = 3 * n - 2$$

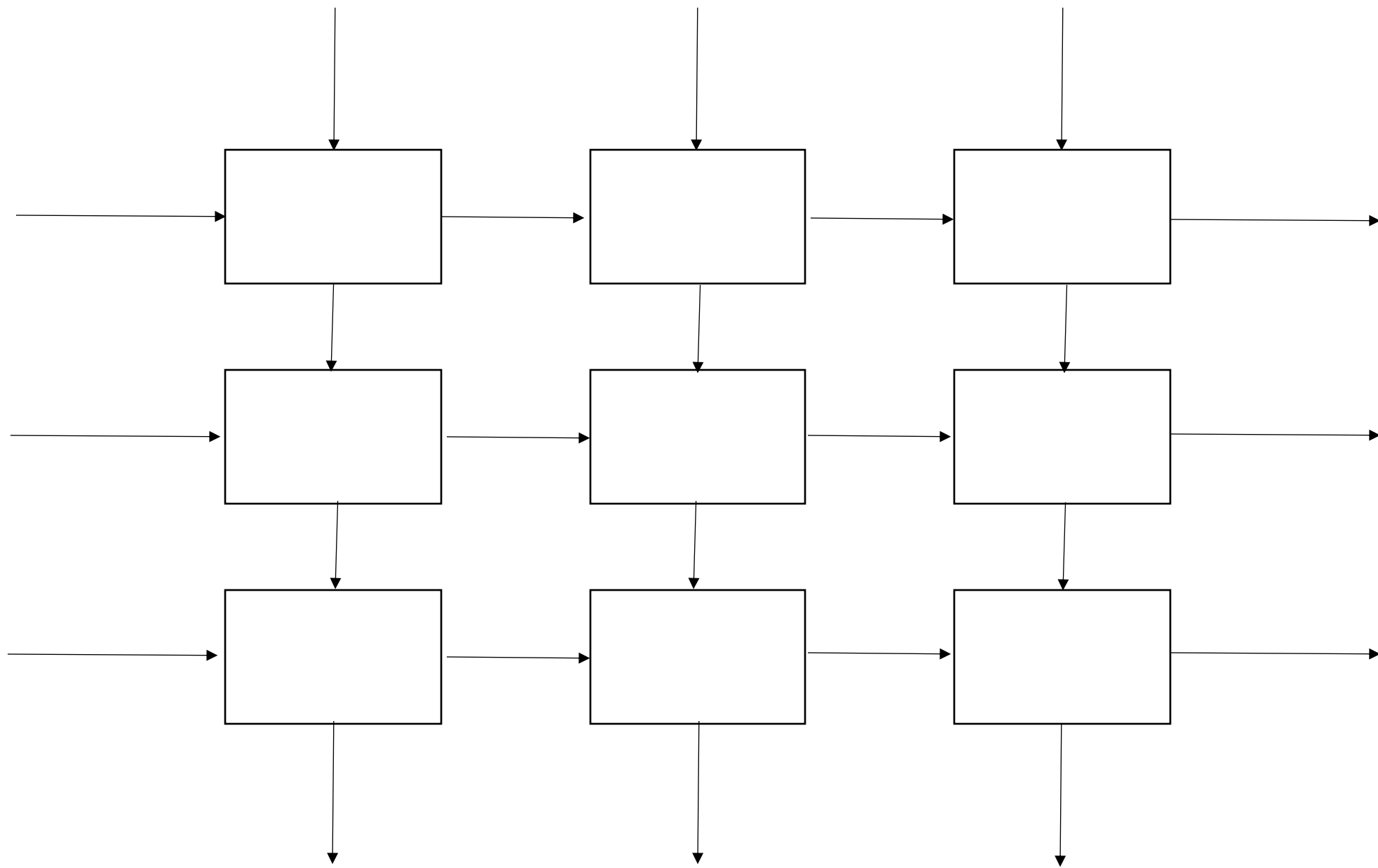
For n x n matrix

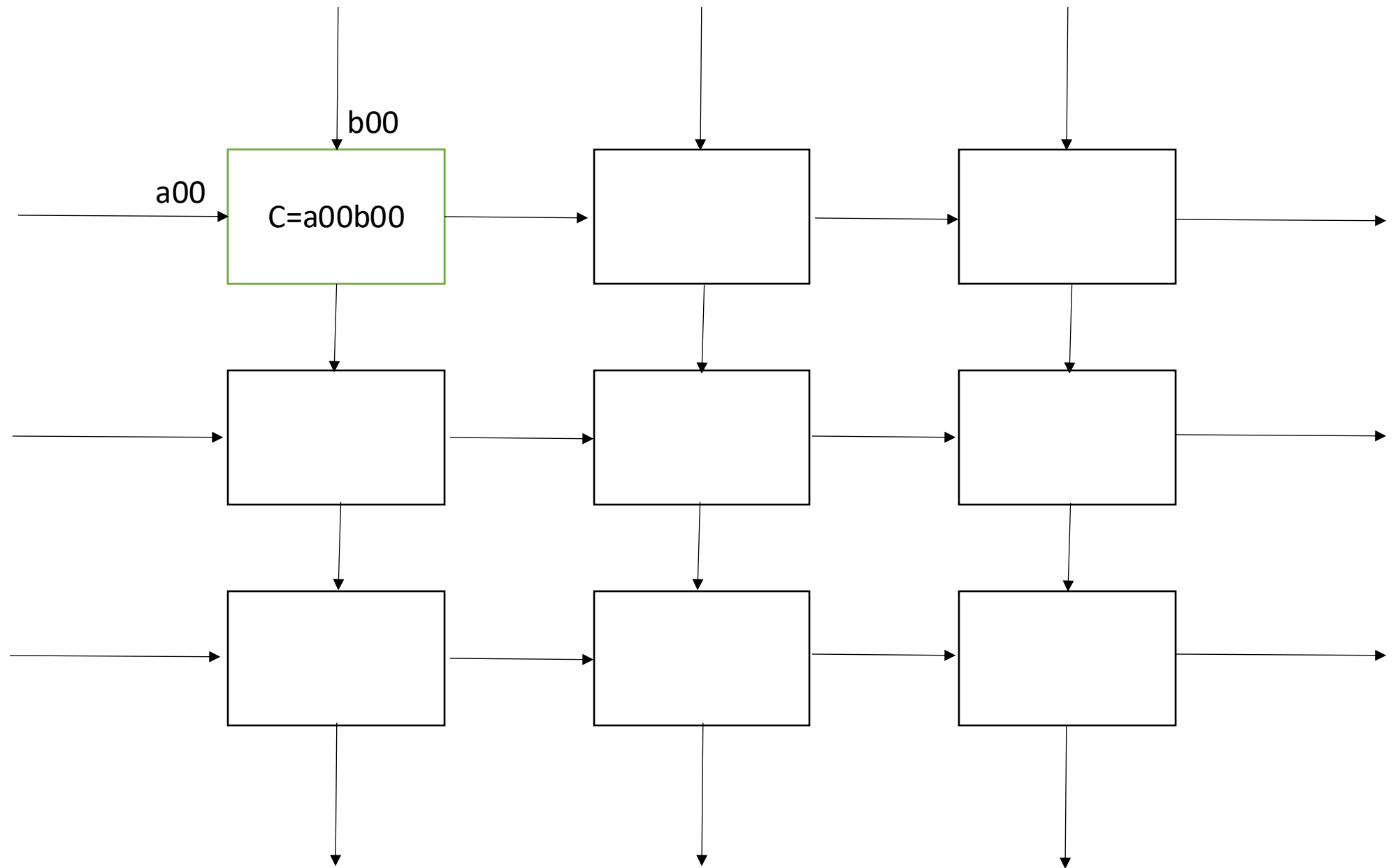
$$c = \begin{matrix} a_{00}b_{00} + a_{01}b_{10} + a_{02}b_{20} & a_{00}b_{01} + a_{01}b_{11} + a_{02}b_{21} & a_{00}b_{02} + a_{01}b_{12} + a_{02}b_{22} \\ a_{10}b_{00} + a_{11}b_{10} + a_{12}b_{20} & a_{10}b_{01} + a_{11}b_{11} + a_{12}b_{21} & a_{10}b_{02} + a_{11}b_{12} + a_{12}b_{22} \\ a_{20}b_{00} + a_{21}b_{10} + a_{22}b_{20} & a_{20}b_{01} + a_{21}b_{11} + a_{22}b_{21} & a_{20}b_{02} + a_{21}b_{12} + a_{22}b_{22} \end{matrix}$$

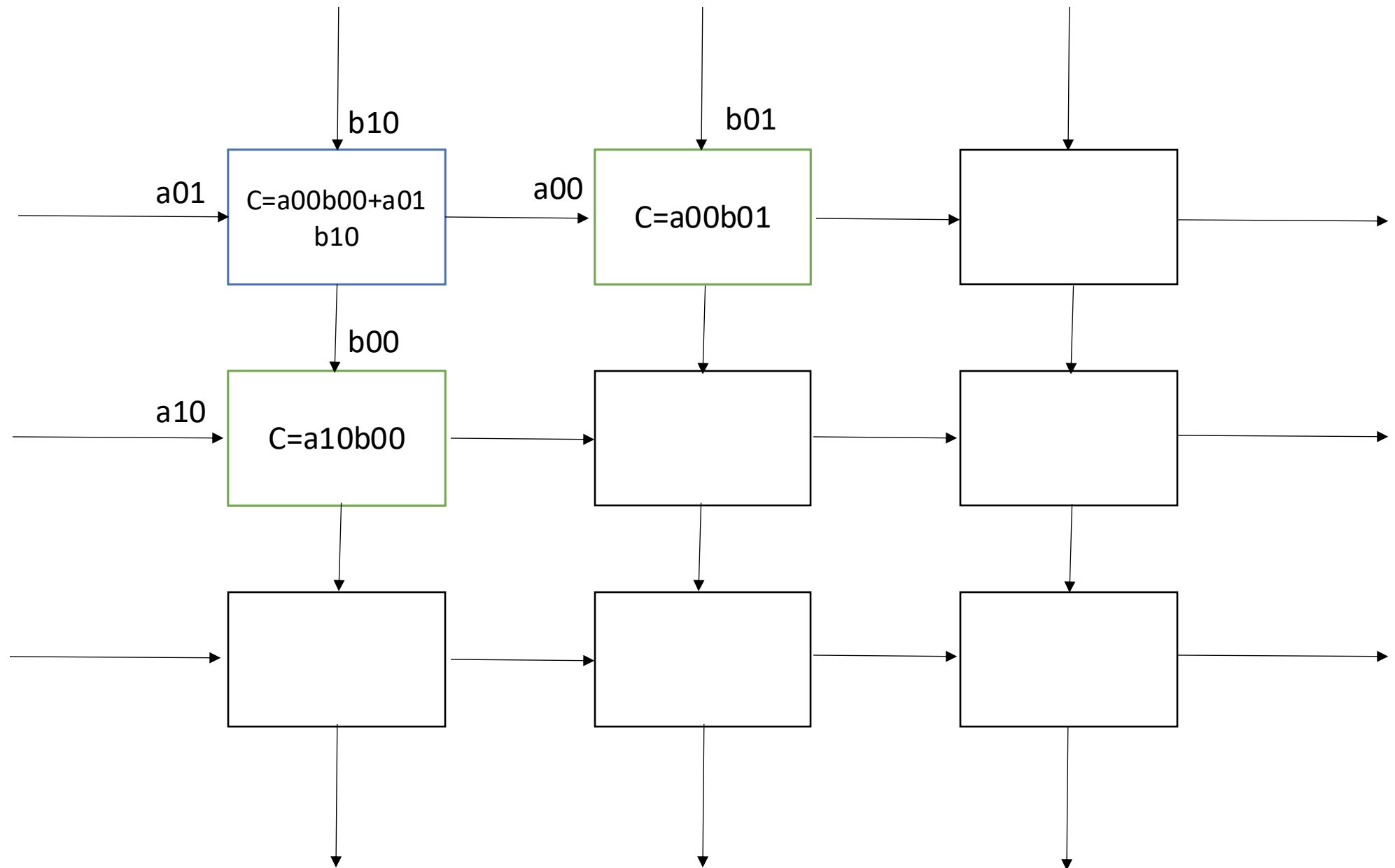
$$\begin{array}{ccc}
 & a_{00} & a_{01} & a_{02} \\
 \bullet \ A = & a_{10} & a_{11} & a_{12} \\
 & a_{20} & a_{21} & a_{22}
 \end{array}
 \quad \longrightarrow \quad
 \begin{array}{ccccc}
 & & & a_{02} & a_{01} & a_{00} \\
 & & a_{12} & a_{11} & a_{10} & \\
 a_{22} & a_{21} & a_{20} & & &
 \end{array}$$

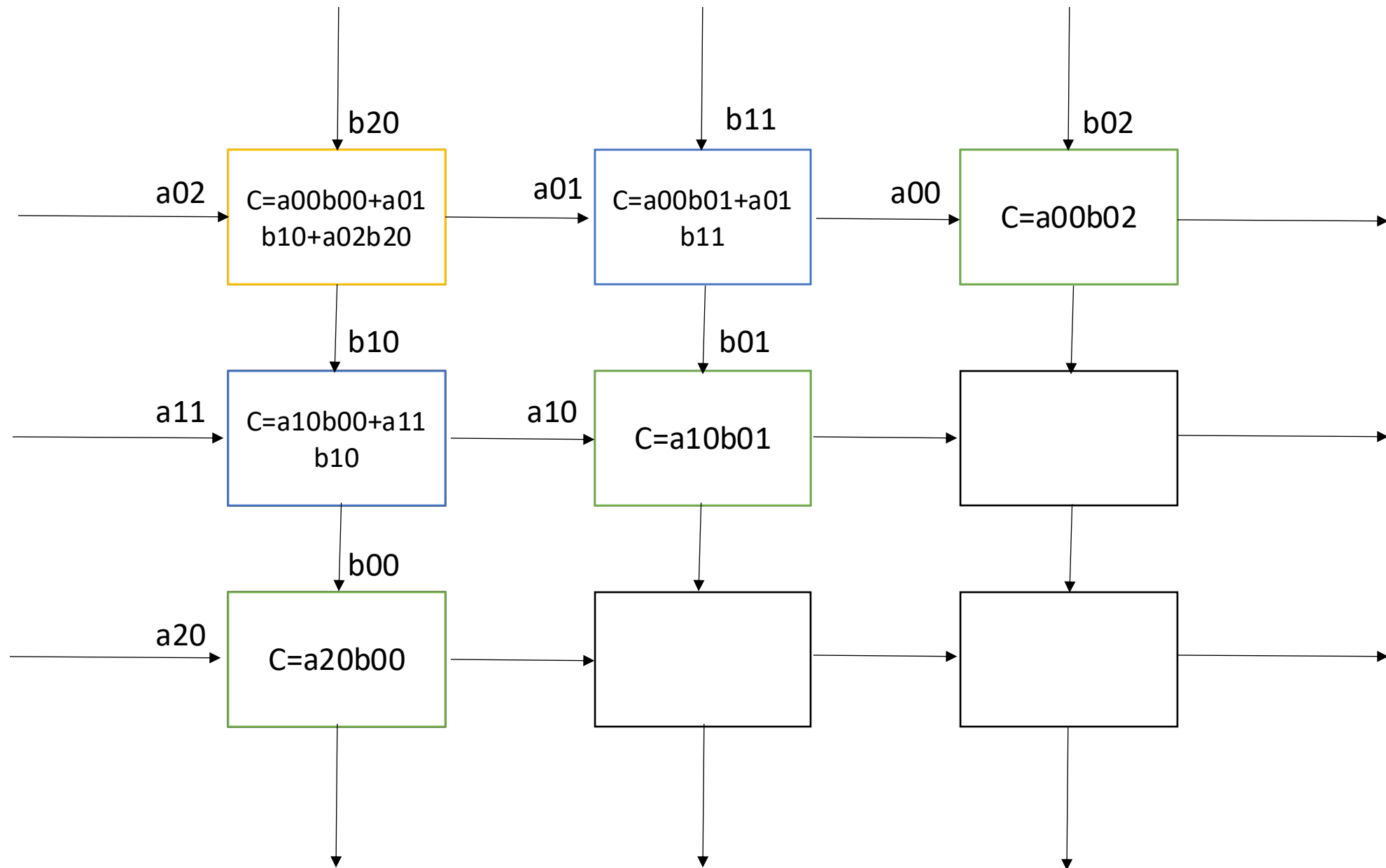
$$\begin{array}{l}
 \bullet \text{ B} = \begin{array}{ccc} b_{00} & b_{01} & b_{02} \\ b_{10} & b_{11} & b_{12} \\ b_{20} & b_{21} & b_{22} \end{array} \quad \longrightarrow \quad \begin{array}{ccc} & & b_{22} \\ & b_{21} & b_{12} \\ b_{20} & b_{11} & b_{02} \\ b_{10} & b_{01} & \\ b_{00} & & \end{array}
 \end{array}$$

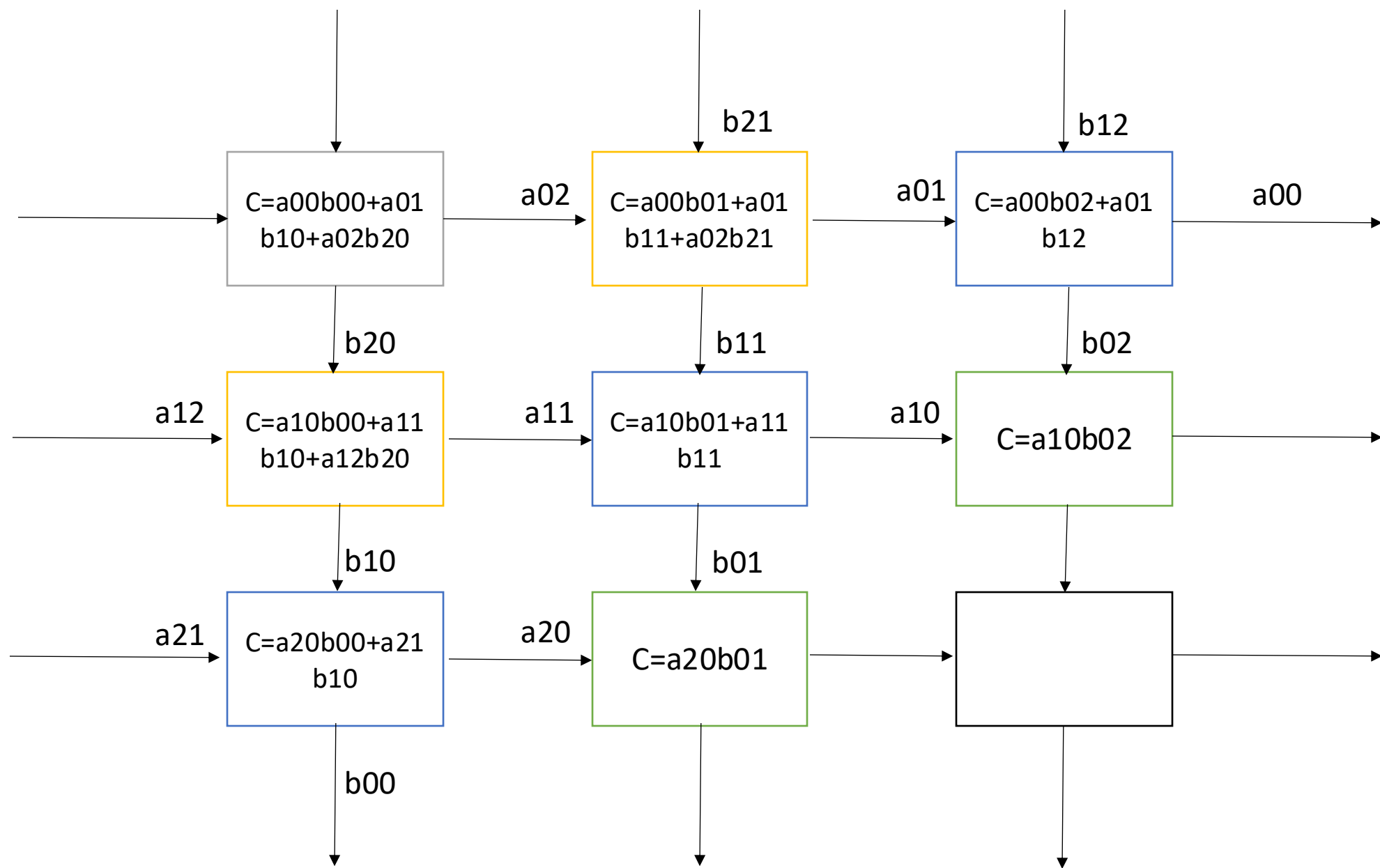


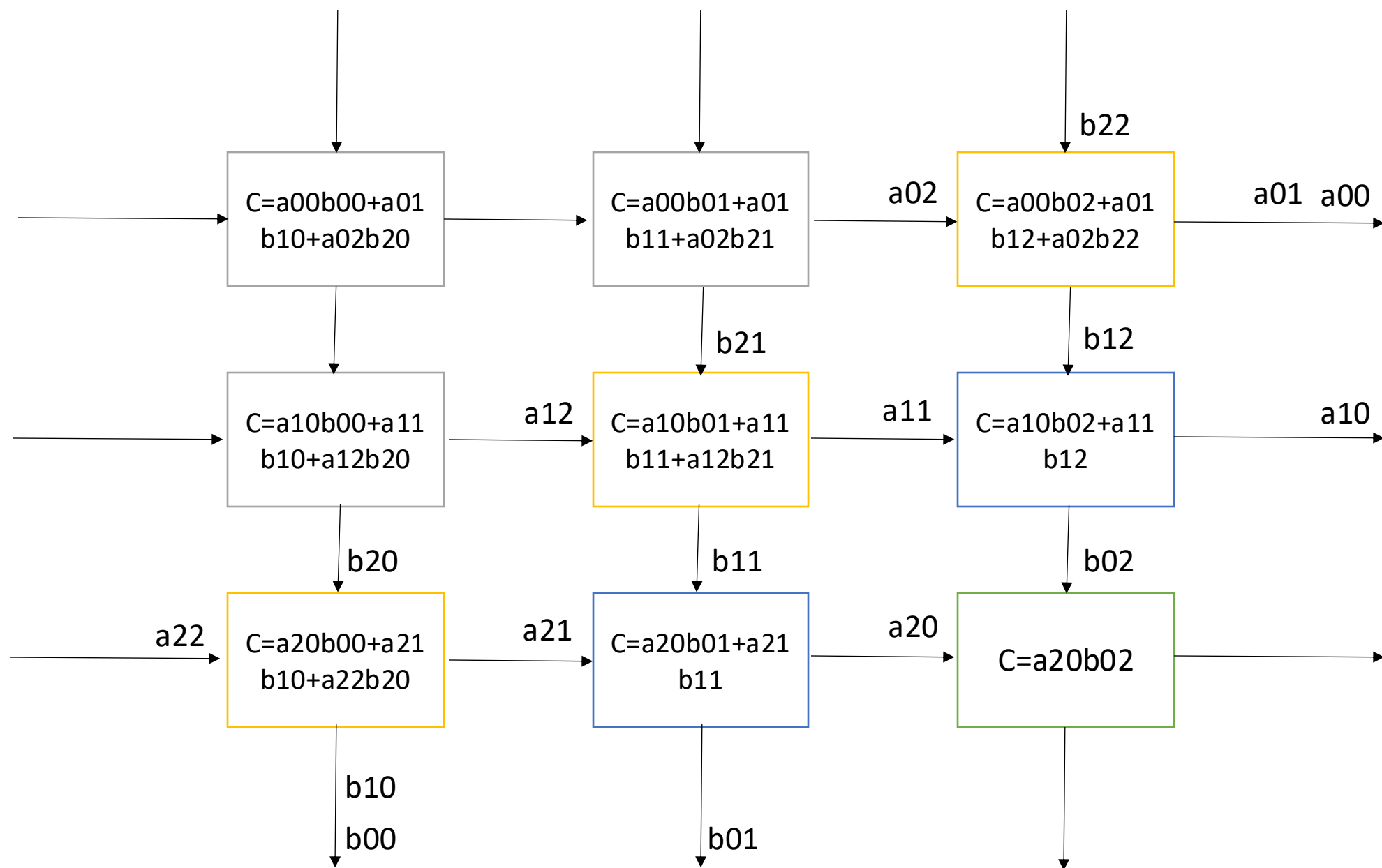


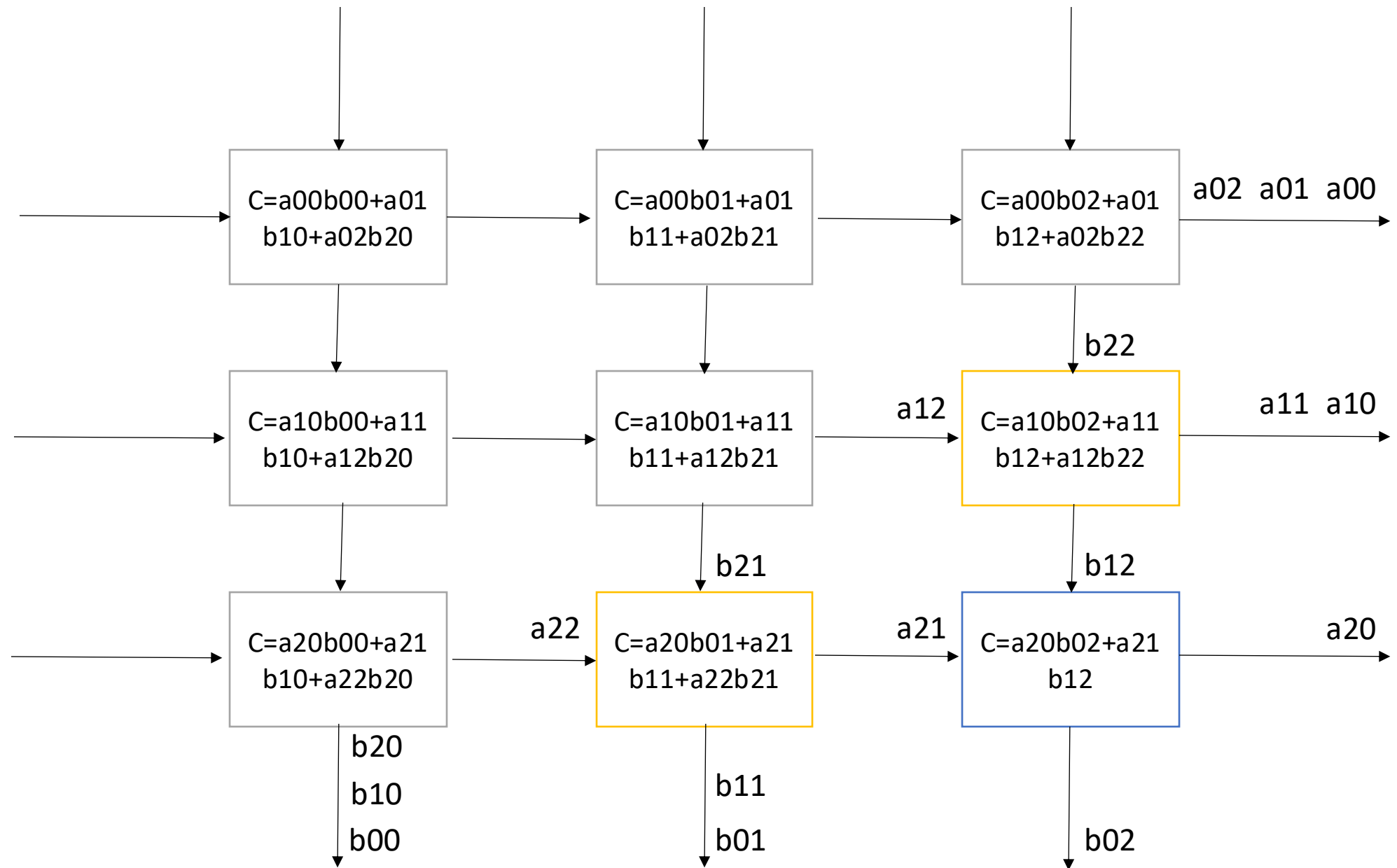


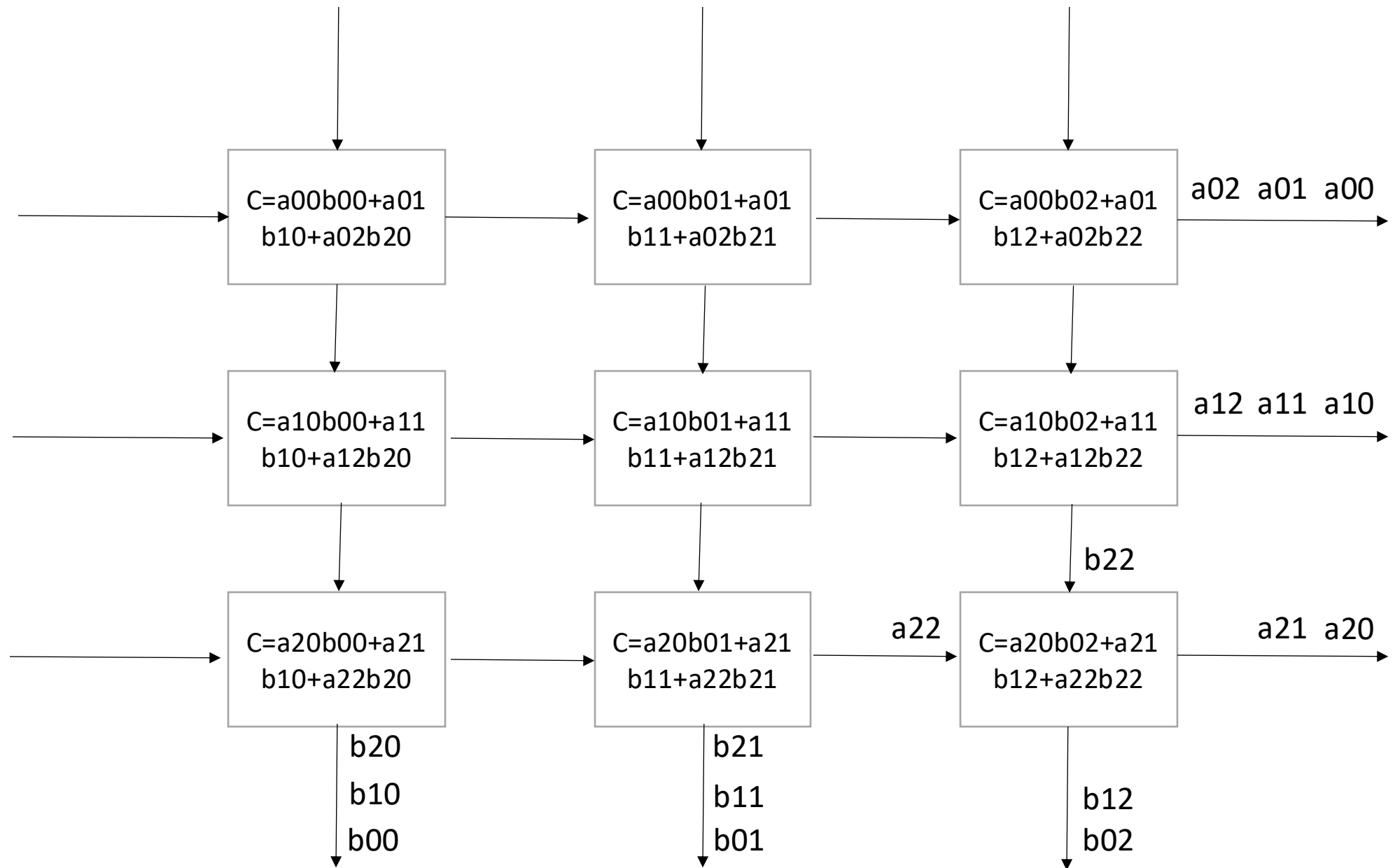












Implementation of Multiplication Module

Multiplication logic implemented using four modules.

- 1) PE module
- 2) Matrix_multiplication
- 3) SimplifiedMasterController
- 4) SystolicMatrixMultiplier

Multiplying array system

PE module

This Module is to define a single PE unit which performs the function to multiply and accumulate, and carry the output to the next PE modules

Matrix_multiplication

This module SystolicMatrixMultiplier coordinates matrix multiplication using a systolic array architecture. It connects a controller and computation unit to process 3×3 matrix inputs and produce a 3×3 output matrix

Controlling system

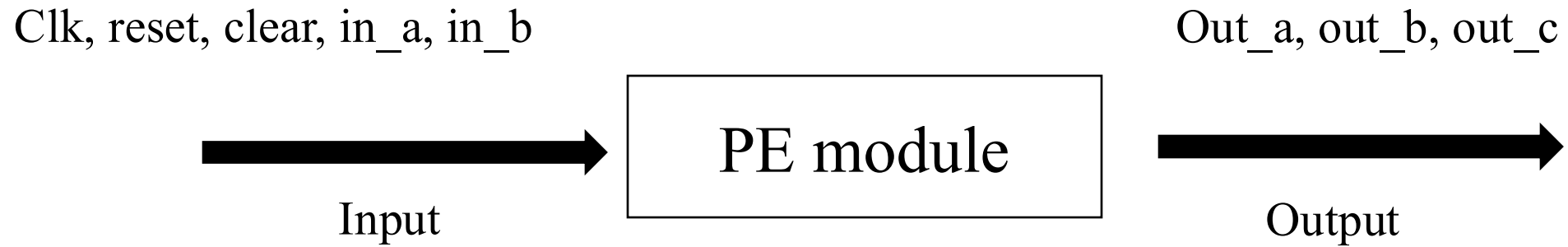
SimplifiedMasterController

The SimplifiedMasterController module controls the timing and flow of matrix data into the systolic array. It simulates the clock-by-clock feeding of matrix elements by sending appropriate values of matrices A and B into the array across multiple cycles, clearing the array initially and signaling completion with a done flag at the end of the data transfer process.

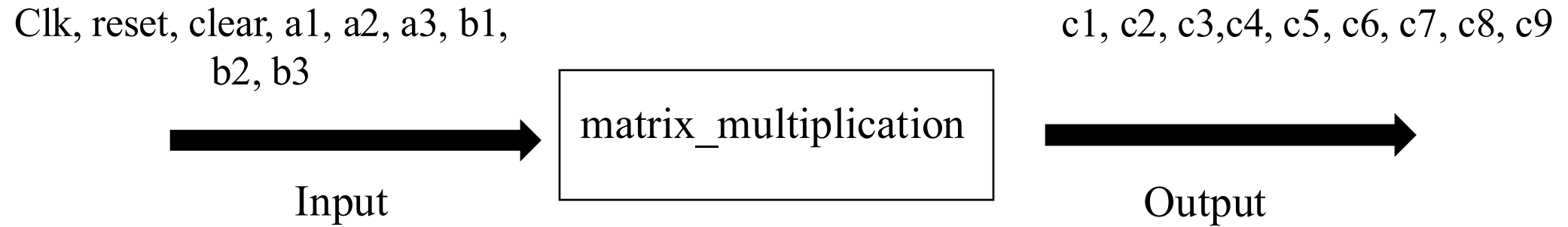
SystolicMatrixMultiplier

The SystolicMatrixMultiplier is the main module that controls the entire matrix multiplication process. It coordinates between the controller and the systolic array to perform 3×3 matrix multiplication and outputs the final result along with a done signal.

Input / Output for Multiplying Modules



Input / Output for Multiplying Modules



Input / Output for Multiplying Modules

Clk, reset, clear, A00, A01, A02, A10, A11,
A12, A20, A21, A22, B00, B01, B02, B10,
B11, B12, B20, B21, B22



Input

SystolicMatrixMultiplier

C00, C01, C02, C10, C11, C12, C20,
C21, C22



Output

Input / Output for Multiplying Modules

Clk, reset, clear, A00, A01, A02, A10, A11,
A12, A20, A21, A22, B00, B01, B02, B10,
B11, B12, B20, B21, B22



Input

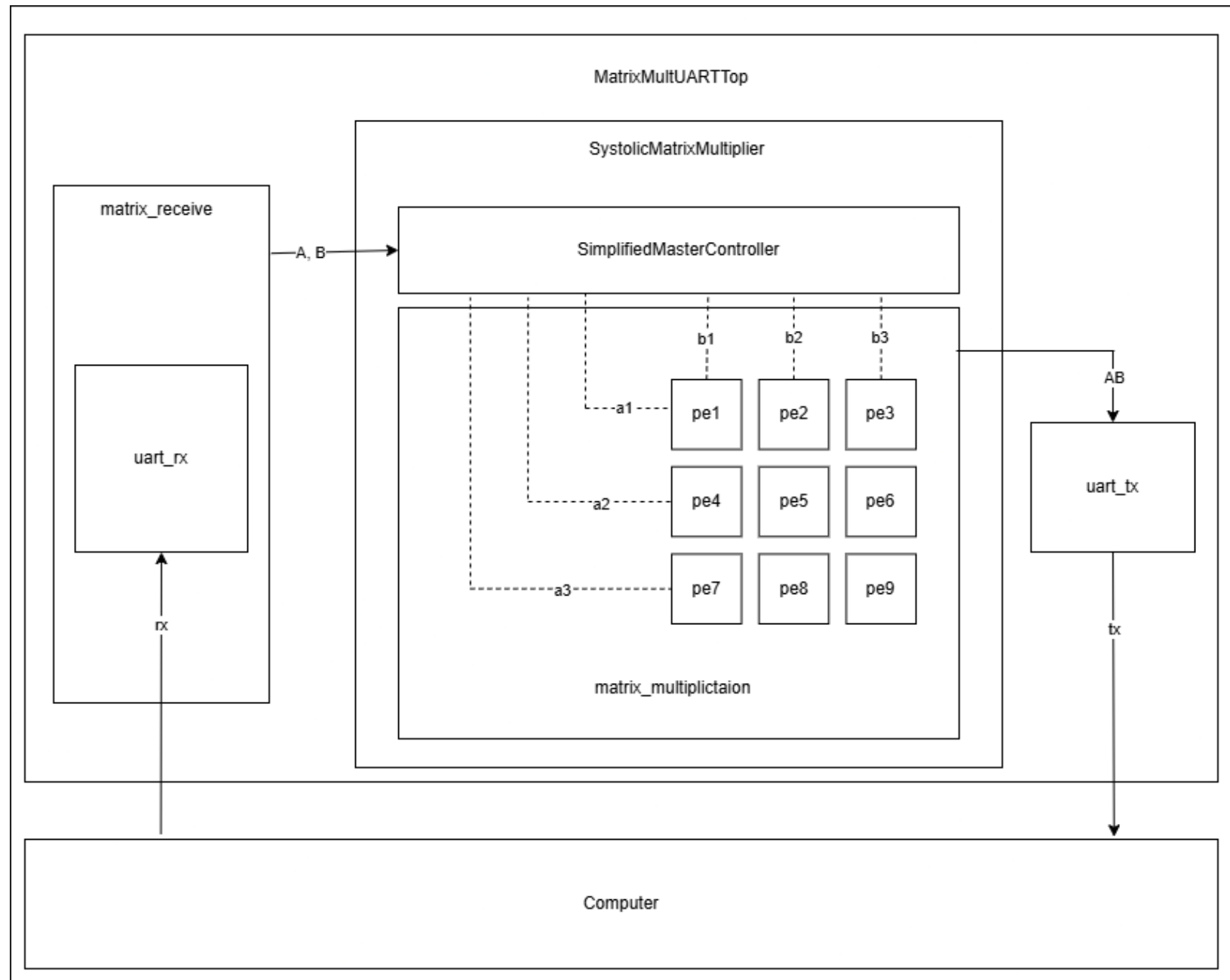
SimplifiedMasterController



a1, a2, a3, b1, b2, b3, done, clear

Output

- Block diagram



UART base Modules

- 1. `uart_rx` :** This module reads rx line and based on baud rate, clock and oversampling rate returns an 8-bit number using majority voting and oversampling.
- 2. `uart_tx` :** This module takes an 8-bit number as input and based on the baud rate and clock, toggles tx for transmission of this number after receiving a high signal in input tx_start.

Controlling and Top Modules

- 1. matrix_receive** : Using the uart_rx module it reads and outputs the 18 input numbers (9 from each input matrix) from the rx line.
- 2. MatrixMultUARTTop** : This is the top module integrating the UART architecture with the SystolicMatrixMultiplier module. It takes rx as input from which it reads all the inputs and after multiplication, returns each element of the output matrix in 3 byte representation on tx.

Error/Problems faced in Multiplications

To pass values to the Processing Elements (PEs), we considered two approaches:

- Store all input values initially, and then update the variables a_1 , a_2 , a_3 , b_1 , b_2 , b_3 after each cycle (i.e., after each round of passing values to the PE's).
- Process inputs incrementally by accepting one set of values, running a cycle, and then repeating this process until all input values are exhausted.

We chose the first approach, as in real-world scenarios, input values are typically made available all at once rather than being passed incrementally.

Unique Approaches

- We broke down our code into many modules which perform different functions and dependent on each other for task. We also made use of MasterController and Simplified MasterController modules to control and update flow of data.

Error/Problems faced in UART

- Our UART tx output was skipping even indexed outputs.
- When we were expecting the outputs to be A, B, C and D, we received outputs A, A, C and C.
- It was caused due to flags being turned on for more cycles than required causing overlap of flags. Solved this by adding buffer states.

T_ASSERT: Assert tx_start for one cycle.

T_WAIT_HIGH: Wait for tx_busy to become high.

T_WAIT_LOW: wait for tx_busy to become low.

Error/Problems faced in UART

- We faced similar errors in matrix_receive module too.
- The rx_done signal was being turned on for more than 1 cycles and should have been detected only once.
- Thus, we changed the code to detect positive edge in this signal by storing its value from the previous cycle in rx_done_d.

Implementation on FPGA

- [Recordings Links](#)
- Note: To see the rx and tx LED flickering on the FPGA board, see the slow motion recording.

References

- [UART implementation by Shubhayu Das](#)
- [Matrix Multiplication using Systolic Arrays, video](#)
- [Matrix Multiplication on a Weight-Stationary Systolic Architecture](#)
- [Systolic Arrays PPT Animation](#)